

TP1 Test Cases

Team 4 CSE 360 TP1 Covers: New Username Recognizer, Password Recognizer, Email Address Validator, application-level field validation, and the Phase 1 functional user stories (US-01 through US-08)

1. Username Recognizer

State machine summary: Start → State 1 requires an ALChar (A–Z, a–z). From State 1, UNChar (A–Z, a–z, 0–9) loops back to State 1. A special character (-, _, ., &) moves State 1 → State 2, and State 2 requires a UNChar to return to State 1. A valid username must end in State 1 (i.e., never end on a special character, and never have two special characters in a row).

ID	Input	Expected Result	Rule Being Tested
TC-U01	abc	Valid	Letters only
TC-U02	abc123	Valid	Letters followed by digits
TC-U03	a1b2c3	Valid	Mixed alphanumeric
TC-U04	user-name	Valid	Single dash between two UNChars
TC-U05	user_name	Valid	Single underscore between two UNChars
TC-U06	user.name	Valid	Single period between two UNChars
TC-U07	user&name	Valid	Single ampersand between two UNChars
TC-U08	a	Invalid	Minimal one-letter username
TC-U09	1username	Invalid	Must start with ALChar, not a digit
TC-U10	-username	Invalid	Must start with ALChar, not a special character
TC-U11	user--name	Invalid	Two consecutive special characters not allowed
TC-U12	user.-name	Invalid	Two different special characters back-to-back not allowed
TC-U13	username-	Invalid	Cannot end on a special character (never returns to accepting state)
TC-U14	"" (empty string)	Invalid	No initial ALChar to consume
TC-U15	user name	Invalid	Space is not ALChar, UNChar, or an allowed special character
TC-U16	user@name	Invalid	@ is not an allowed special character for usernames

2. Password Recognizer

State machine summary: Each input character is classified as upperCase (A–Z), lowerCase (a–z), numericChar (0–9), specialChar (one of ~ ` ! @ # \$ % ^ & * () _ - + { } [] \ : , . ? /), or otherChar (anything else). charCounter increments per character. On full input consumption: longEnough = True if $8 \leq \text{charCounter} \leq 32$; tooLong = True if charCounter > 32.

ID	Input	Expected Result	Rule Being Tested
TC-P01	Passw0rd! (9 chars)	Valid — upperCase, lowerCase, numericChar, specialChar all True; longEnough True	All four character classes present, valid length
TC-P02	8-character password, e.g. Aa1!Aa1!	longEnough True	Lower boundary of length range
TC-P03	7-character password, e.g. Aa1!Aa1	longEnough False	Just below minimum length
TC-P04	32-character password with mixed classes	longEnough True	Upper boundary of length range

ID	Input	Expected Result	Rule Being Tested
TC-P05	33-character password with mixed classes	tooLong True	Just above maximum recognizer length
TC-P06	"" (empty string)	charCounter = 0; all flags False; longEnough False	No input consumed
TC-P07	alllowercase	lowerCase True only (assuming length valid)	Only-lowercase input
TC-P08	ALLUPPERCASE	upperCase True only	Only-uppercase input
TC-P09	12345678	numericChar True only	Only-digit input
TC-P10	!@#\$\$%^&*	specialChar True only	Only allowed-special-character input
TC-P11	Password! (no digit)	numericChar False; upperCase, lowerCase, specialChar True	Missing one character class (feedback check for US-06)
TC-P12	Password containing é or a control character	otherChar True	Character outside all four defined classes
TC-P13	Password containing ; or '	otherChar True	Character not in the defined special-character list
TC-P14	Valid password with all 4 classes, length exactly 8	All relevant flags True, longEnough True, tooLong False	Combined boundary + completeness case

3. Email Address Validator

State machine summary: Local part: State 0 → State 1 on LPChar; State 1 loops on LPChar; a . moves State 1 → State 0, which requires another LPChar to return to State 1 (so a period can't lead/trail or repeat in the local part). @ moves State 1 → State 2. Domain part: State 2 → State 3 on DPChar; State 3 loops on DPChar; . moves State 3 → State 2 (needs another DPChar to return); - moves State 3 → State 4 (needs another DPChar to return to State 3). EoI (end of input) is only accepted from State 3.

ID	Input	Expected Result	Rule Being Tested
TC-E01	user@example.com	Valid	Basic well-formed address
TC-E02	first.last@example.com	Valid	Single period inside local part
TC-E03	user@my-site.com	Valid	Single dash inside domain
TC-E04	user@mail.example.co.uk	Valid	Multiple domain labels separated by periods
TC-E05	a@b.co	Valid	Minimal-length local part and domain
TC-E06	userexample.com	Invalid	Missing @
TC-E07	.user@example.com	Invalid	Local part cannot start with .
TC-E08	user..name@example.com	Invalid	Consecutive periods not allowed in local part
TC-E09	user.@example.com	Invalid	Local part cannot end on . before @
TC-E10	@example.com	Invalid	Local part cannot be empty (no LPChar before @)
TC-E11	user@	Invalid	Domain part cannot be empty
TC-E12	user@.com	Invalid	Domain cannot start with .
TC-E13	user@example.com.	Invalid	Domain cannot end on . (EoI only valid from State 3)
TC-E14	user@example.com-	Invalid	Domain cannot end on -
TC-E15	user@exa--mple.com	Invalid	Consecutive dashes not allowed in domain
TC-E16	user@example..com	Invalid	Consecutive periods not allowed in domain
TC-E17	us er@example.com	Invalid	Space is not a valid LPChar
TC-E18	user@exam ple.com	Invalid	Space is not a valid DPChar

ID	Input	Expected Result	Rule Being Tested
TC-E19	Big Email	Invalid	Too many characters

4. Functional / User Story Test Cases

These tests validate the Phase 1 user stories themselves (TP1 Plan.pdf / Sprint Backlog.pdf), beyond field-level validation. IDs here are functionality-based (TC-F##) rather than tied to a specific US-number.

4.1 One-time password reset

ID	Scenario	Expected Result	Story Tested
TC-F01	Admin sets a one-time password for a user who forgot their password	User's password is updated to the one-time value and flagged as one-time	Admin can issue a reset
TC-F02	User logs in using a valid one-time password	Login succeeds; user is immediately required to set a new password before reaching the application	Forced password change is enforced
TC-F03	After the new password is set, the same one-time password is used again to log in	Invalid — login rejected	One-time password is cleared after use, cannot be reused
TC-F04	User finishes setting the new password	User is returned to the login screen and must log in again with the new password	Re-login is required after reset

4.2 Delete user account

ID	Scenario	Expected Result	Story Tested
TC-F05	Admin selects a user account and chooses Delete	A confirmation prompt ("Are you sure?") is shown before anything is deleted	Delete requires confirmation
TC-F06	Admin answers "Yes" to the confirmation prompt	Account is removed; that user can no longer log in	Confirmed deletion takes effect
TC-F07	Admin declines / closes the confirmation prompt	Account is not deleted	Cancelling the confirmation preserves the account
TC-F08	Admin attempts to delete their own currently logged-in account	Invalid — action blocked	Self-deletion is prevented

4.3 List all user accounts

ID	Scenario	Expected Result	Story Tested
TC-F09	Admin opens the account listing screen	Every account is listed with username, full name, email address, and assigned role(s)	Admin can list all accounts
TC-F10	A new account is created, then the listing is refreshed	The new account appears in the list with correct details	Listing reflects current account data

4.4 Add / remove roles per user

ID	Scenario	Expected Result	Story Tested
TC-F11	Admin adds a role to an existing user	The role appears in that user's assigned roles and the user gains the associated access	Admin can add a role

ID	Scenario	Expected Result	Story Tested
TC-F12	Admin removes a role from an existing user	The role no longer appears for that user and associated access is revoked	Admin can remove a role
TC-F13	Admin attempts to remove a user's only remaining role	Team decision needed: whether a user must always retain at least one role	Open item confirm expected behavior with the team before finalizing this test

4.5 Username validation integration

ID	Scenario	Expected Result	Story Tested
TC-F14	New Account screen, username meeting all recognizer and length rules	Account creation proceeds	Username validation is wired into account creation
TC-F15	New Account screen, invalid username (e.g., starts with a digit)	Account creation is blocked and a specific message identifies which rule failed	Clear, rule-specific error messaging on failure

4.6 Password live feedback

ID	Scenario	Expected Result	Story Tested
TC-F16	User begins typing a new password	UI updates in real time to show which requirements (uppercase, lowercase, digit, special character, length) are currently satisfied	Live feedback during password entry
TC-F17	User types a password missing one character class (e.g., no digit)	The indicator for that class remains unsatisfied while the others show satisfied	Feedback is accurate per class, not just pass/fail overall
TC-F18	User's password reaches the maximum-length boundary	Feedback reflects the length requirement, including a "too long" indicator not present in the original state diagram	Maximum-length feedback, added beyond the base state diagram

4.7 Replace placeholder roles (Role1 / Role2)

ID	Scenario	Expected Result	Story Tested
TC-F19	Inspect every screen and code path where Role1 / Role2 previously appeared	No placeholder role names remain visible anywhere in the running application	Placeholder roles fully replaced
TC-F20	Assign one of the new real roles to a user	Role behaves consistently with the existing role-handling logic (login, access, listing)	New roles follow the same pattern as prior roles